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Methods of data Modeling

When a data reveals a certain trend it could be statistically estimated with the help of an equation. The process of determining the equation that fits the given data is known as curve fitting or model fitting of data.

Some curve fitting models are:

- Linear model (simple regression equation)
- Multiple regression model
- Second degree equation (parabolic curve)
- Exponential curve

(A) Simple regression model

The simple regression model is used to find the value of unknown (dependent) variable with the help of given or known or independent variable. The equation of simple regression model is $Y = a + bx$ — (I). To find the value of 'a' and 'b', we develop two normal equations with the help of eqn (I) i.e.

$$\sum Y = na + b \sum x \quad \text{--- (II)}$$

$$\sum XY = a \sum x + b \sum x^2 \quad \text{--- (III)}$$

Solving eqn (II) & (III) we get the value of a and b then substituting in eqn (i) we get the required line of regression.

A panel of two judges A and B graded 7 dramatic performance by independently awarding marks as follows:

Performance	1	2	3	4	5	6	7
Marks by A(x)	46	42	44	40	43	41	45
Marks by B(y)	40	38	36	35	39	37	41

- (I) Fit the straight line trend of type $Y = a + bX$
 (II) Estimate the value of Y when $X = 43$

Solution:

X	Y	X^2	XY
46	40	2116	1840
42	38	1764	1596
44	36	1936	1584
40	35	1600	1400
43	39	1849	1677
41	37	1681	1517
45	41	2025	1845

Total 301 266 12971 11459

The equation of trend line is:

$$Y = a + bX \quad \text{--- (I)}$$

The two normal equations for eqn (I) are:

$$\sum Y = na + b \sum X \quad \text{--- (II)}$$

$$\sum XY = a \sum X + b \sum X^2 \quad \text{--- (III)}$$

Now substituting the values of eqn (ii) and (iii)

$$7a + 301b = 266 \quad \text{--- (iv)}$$

$$301a + 12971b = 11459 \quad \text{(v)}$$

Now, solving eqn (iv) and (v):

Multiplying (iv) by 43 Then:

$$301a + 12943b = 11438$$

$$\underline{301a + 12971b = 11459}$$

$$-28b = -21$$

$$\therefore b = \frac{21}{28} = 0.75$$

Again: Then eqn (iv):

$$7a + 301b = 266$$

$$\text{or, } 7a + 301 \times 0.75 = 266$$

$$\therefore a = 5.75$$

Now, substituting the value of a and b in eqn (i):

$$Y = a + bx$$

$$\therefore Y = 5.75 + 0.75x$$

⑥ When $x = 43$

Then: most likely value of Y is:

$$Y = 5.75 + 0.75 \times 43 = 38$$

⑧ Multiple regression equation:

The regression equation is used to estimate the value of unknown (dependent) variable with the help of known (independent) variable. The multiple regression equation is used to estimate the value of unknown (dependent) variable with the help of two or more independent (known) variable. Consider the three variables Y , X_1 and X_2 then the regression equation of Y (dependent variable) on X_1 and X_2 (independent variable) is given as:

$$Y = a + b_1 X_1 + b_2 X_2 \quad \dots (i)$$

The three normal equations for eqn (i) are:

$$\sum Y = na + b_1 \sum X_1 + b_2 \sum X_2 \quad \dots (ii)$$

$$\sum YX_1 = a \sum X_1 + b_1 \sum X_1^2 + b_2 \sum X_1 X_2 \quad \dots (iii)$$

$$\sum YX_2 = a \sum X_2 + b_1 \sum X_1 X_2 + b_2 \sum X_2^2 \quad \dots (iv)$$

Solving eqn (ii), (iii) and (iv) we get the value of a , b_1 and b_2 .

Then, substituting values of a , b_1 and b_2 in eqn (i) we get the required multiple regression equation.

A survey on expenditure of some families resulted the following data.

Expenditure on food (Rs 1000) (Y)	Annual income (Rs. 1000) (X_1)	Family size (X_2)
--	---------------------------------------	--------------------------

Y	X_1	X_2
7	30	3
9	45	2
10	35	4
11	55	5
13	30	1

Estimate the expenditure on food of a family with annual income of Rs. 60,000 and having 4 family members.

Soln

The regression eqn of exp. (Y) on annual income (X_1) and family size (X_2) is:

$$Y = a + b_1 X_1 + b_2 X_2 \quad \text{--- (i)}$$

The three normal equations for eqn (i) are:

$$\sum Y = na + b_1 \sum X_1 + b_2 \sum X_2 \quad \text{--- (ii)}$$

$$\sum YX_1 = a \sum X_1 + b_1 \sum X_1^2 + b_2 \sum X_1 X_2 \quad \text{--- (iii)}$$

$$\sum YX_2 = a \sum X_2 + b_1 \sum X_1 X_2 + b_2 \sum X_2^2 \quad \text{--- (iv)}$$

X_1	X_2	Y	YX_1	YX_2	X_1X_2	X_1^2	X_2^2
30	3	7	210	21	90	900	9
45	2	9	405	18	90	2025	4
35	4	10	350	40	140	1225	16
55	5	11	605	55	275	3025	25
30	1	13	390	13	30	900	1
Total	195	50	1960	147	625	8075	55

Substituting the values in eqn (ii), (iii) & (iv)

$$5a + 195b_1 + 15b_2 = 50 \quad \text{--- (v)}$$

$$195a + 8075b_1 + 625b_2 = 1960 \quad \text{--- (vi)}$$

$$15a + 625b_1 + 55b_2 = 147 \quad \text{--- (vii)}$$

Solving eqn (iv), (v), (vi) & (vii) we have:

Multiplying eqn (v) by 39. Then;

$$1950 + 7605b_1 + 585b_2 = 1950$$

$$1950 + 8075b_1 + 625b_2 = 1960$$

$$-470b_1 - 40b_2 = -10$$

$$-(470b_1 + 40b_2) = -10$$

$$\therefore 470b_1 + 40b_2 = 10 \quad \dots \dots (viii)$$

Multiplying eqn (v) by 3. Then;

$$150 + 585b_1 + 45b_2 = 150$$

$$150 + 625b_1 + 55b_2 = 147$$

$$+40b_1 + 10b_2 = +3$$

$$-(40b_1 + 10b_2) = +3$$

$$\therefore -(40b_1 + 10b_2) = 3 \quad \dots \dots (ix)$$

Solving eqn (viii) and (ix):

$$470b_1 = 10 - 40b_2$$

$$\therefore b_1 = \frac{10 - 40b_2}{470} \quad \dots \dots (x)$$

Then;

$$-40 \times \left(\frac{10 - 40b_2}{470} \right) + 10b_2 = 3$$

$$\text{or, } -400 + 1600b_2 + 4700b_2 = 1410$$

$$\text{or, } -1600b_2 = -3690 \quad -1810 = 3100b_2$$

$$\therefore b_2 = -0.584$$

$$a = 8.93$$

$$b_1 = 0.071$$

$$b_2 = -0.584$$

Substituting the values of a , b_1 and b_2 in eqn (i)
i.e. $Y = a + b_1 X_1 + b_2 X_2$

$$\text{or, } Y = 8.93 + 0.071 \times X_1 - 0.584 X_2$$

Here,

Annual income (X_1) = 60

family size (X_2) = 4

Then expected expenditure of food is:

$$\begin{aligned} Y &= 8.93 + 0.071 X_1 - 0.584 X_2 \\ &= 8.93 + 0.071 \times 60 - 0.584 \times 4 \\ &= 8.93 + 4.26 - 2.336 \\ &= 13.19 - 2.336 \\ &= 10.584, \end{aligned}$$

(C) Second degree curve (Parabolic curve)

The mathematical equation of parabolic curve is:

$$y = a + b_1 x + b_2 x^2 \quad \text{--- (I)}$$

The three normal equations for eqn (I) are:

$$\sum Y = na + b_1 \sum X + b_2 \sum X^2 \quad \text{--- (II)}$$

$$\sum XY = a \sum X + b_1 \sum X^2 + b_2 \sum X^3 \quad \text{--- (III)}$$

$$\sum X^2 Y = a \sum X^2 + b_1 \sum X^3 + b_2 \sum X^4 \quad \text{--- (IV)}$$

Solving eqn (II), (III) and (IV) we get;

The value of a , b_1 & b_2 .

Fit the second degree parabola for the following data:

X	1	2	3	4	5	6
Y	6	17	34	57	86	121

Soln:

The mathematical equation of parabolic curve is:

$$Y = a + b_1x + b_2x^2 \quad \dots (i)$$

The three normal equations for eqn (i) are:

$$\sum Y = na + b_1 \sum x + b_2 \sum x^2 \quad \dots (ii)$$

$$\sum XY = a \sum x + b_1 \sum x^2 + b_2 \sum x^3 \quad \dots (iii)$$

$$\sum x^2 Y = a \sum x^2 + b_1 \sum x^3 + b_2 \sum x^4 \quad \dots (iv)$$

X	Y	x^2	x^3	x^4	xy	x^2y
1	6	1	1	1	6	6
2	17	4	8	16	34	68
3	34	9	27	81	102	306
4	57	16	64	256	228	912
5	86	25	125	625	430	2150
6	121	36	216	1296	726	4356
21	321	91	441	2275	1526	7798

Substituting value in eqn (ii), (iii) & (iv):

$$6a + 21b_1 + 91b_2 = 321 \quad \dots (v)$$

$$21a + 91b_1 + 441b_2 = 1526 \quad \dots (vi)$$

$$91a + 441b_1 + 2275b_2 = 7798 \quad \dots (vii)$$

ing data:

Solving (v), (vi) and (vii) we get:

$$a=1, b_1=2, b_2=3$$

Now substituting the value of a , b , and b_2 in eqn (i)

$$\text{i.e. } Y = a + b_1x + b_2x^2$$

$$\text{or, } Y = 1 + 2x + 3x^2$$

ve is:

⑥ Exponential Curve

If the value of Y decreasing (or increasing) continuously then the exponential type of curve would be choice.

⑦ Let the curve be $Y = ab^x$ — (I)

Taking log on both sides:

$$\log Y = \log ab^x$$

$$= \log a + \log b^x$$

$$\text{or, } \log Y = \log a + x \log b$$

Let;

$$u = \log Y, \log a = A, \log b = B$$

Then:

$$u = A + Bx \quad \dots (i)$$

Which is the linear equation of u on x .

The two normal equations for eqn (i) are:

$$\sum u = nA + B \sum x \quad \dots (ii)$$

$$\sum ux = A \sum x + B \sum x^2 \quad \dots (iii)$$

Solving eqn (ii) and (iii) we get the value of A and B .

Then; $a = \text{Antilog } A$, $b = \text{Antilog } B$

and substituting the values of a and b in eqn (I) we get required exponential curve.

Fit the exponential curve of the form $Y = ab^x$ to the following data.

X	1	2	3	4	5	6	7	8
Y	1	1.2	1.8	2.5	3.6	4.7	6.6	9.1

Calculations:

The exponential curve be $Y = ab^x$ --- (I)

Taking log on both sides:

$$\log Y = \log ab^x$$

$$\text{or, } \log Y = \log a + x \log b$$

let;

$$u = A + Bx$$

Then; $u = A + Bx$ --- (II)

The normal equations for eqn (II) are:

$$\sum u = nA + B \sum x \quad \text{--- (III)}$$

$$\sum ux = A \sum x + B \sum x^2 \quad \text{--- (IV)}$$

X	Y	$u = \log Y$	xu	x^2
1	1	0	0	1
2	1.2	0.0792	0.1584	4
3	1.8	0.2552	0.7659	9
4	2.5	0.3979	1.5916	16
5	3.6	0.5563	2.7815	25
6	4.7	0.6721	4.0326	36
7	6.6	0.8195	5.7365	49
8	9.1	0.9590	7.6720	64
Total	36	30.5	3.7393	22.7385
				204

Substituting the values in eqn (III) & (IV);

$$8A + 36B = 3.7393 \quad \text{--- (v)}$$

$$36A + 204B = 22.7385 \quad \text{--- (VI)}$$

Solving eqn (v) and (VI) we get;

$$A = -0.166375$$

$$B = 0.1406$$

Then;

$$b = \text{Antilog } B = \text{Antilog } (0.1406) = 1.38$$

$$a = \text{Antilog } A = \text{Antilog } (-0.166375) = 0.68$$

Now, substituting the values of a and b in eqn (I):

$$\text{i.e. } Y = ab^x$$

$$\therefore Y = 0.68 (1.38)^x$$

② Let the curve be $Y = ae^{bx} \quad \text{--- (i)}$

Taking $\ln$ on both sides:

$$\ln Y = \ln ae^{bx}$$

$$\text{or, } \ln Y = \ln a + bx \ln e$$

$$\text{or, } \ln Y = \ln a + bx \quad (\ln e = 1)$$

$$\text{let } y = \ln Y, \quad A = \ln a$$

Then the equation be;

$$y = A + bx \quad \text{--- (ii)}$$

which is the linear regression equation of Y on x .

The normal equations for eqn (ii) are:

$$\sum Y = nA + b \sum x \quad \text{--- (iii)}$$

$$\sum XY = A \sum x + b \sum x^2 \quad \text{--- (iv)}$$

Solving (iii) and (iv) we get value of A and b.
Then, $a = e^A = (\text{Anti } \ln)$

Fit the curve of type $y = ae^{bx}$ for the following data:

Time in day	1	2	3	4	5	6	7	8
No. of bacteria	15	20	27	37	50	65	90	120

Soln;

We have;

$$y = ae^{bx} \quad \dots (i)$$

Taking $\ln$ on both sides:

$$\ln y = \ln ae^{bx}$$

$$\text{or, } \ln y = \ln a + bx \ln e$$

$$\text{or, } \ln y = \ln a + bx$$

$$\text{let, } y = \ln y, A = \ln a$$

$$\text{Then: } y = A + bx \quad \dots (ii)$$

The normal equations for eqn (ii) are:

$$\sum y = nA + b \sum x \quad \dots (iii)$$

$$\sum xy = A \sum x + b \sum x^2 \quad \dots (iv)$$

Calculation:

F.M: 50 Time: 2 hrs
P.M = 17.5

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Date 2078/08/07

Page No.

Tuesday

Group A: Attempt any 6 questions $6 \times 5 = 30$ marks

Group B: Attempt any 2 questions $2 \times 10 = 20$ marks

Time in day (x)	No. of bacteria (y)	$y = \ln y$	xy	x^2
1	15	2.708	2.708	1
2	20	2.995		4
3	27	3.295		9
4	37	3.610		16
5	50	3.912		25
6	65	4.174		36
7	90	4.499		49
8	120	4.787		64
36		29.98 ≈ 30		204

Now, substituting the values in eqn (ii) and (iv):

$$8A + 36B = 30 \quad \text{--- (v)}$$

$$36A + 204B = 147.416 \quad \text{--- (vi)}$$

Solving eqn (v) and (vi) we get: $A = 2.4225$
 $B = 0.2952$

Then; $a = \text{Anti} \ln(A) = e^{2.4225} = 11.2740$

Now; substituting in eqn (i)

$$\text{i.e. } Y = ae^{bx}$$

$$\text{or, } Y = 11.2740e^{0.2952x}$$

Multiple and Partial Correlation

In simple correlation we study the relationship between two variables only. If there are three or more than three variables under the study the correlation may be either multiple or partial.

Multiple Correlation

In multiple correlation we study the combined effect of all independent factors on a dependent factor.

Consider three variables x_1 , x_2 and x_3 .

Then;

$R_{1.23}$ = multiple correlation coefficient with x_1 as dependent and x_2 and x_3 as independent

Similarly;

$R_{2.13}$ = multiple correlation coefficient with x_2 as dependent and x_1 , x_3 as independent variables.

$R_{3.12}$ = multiple correlation coefficient with x_3 as dependent and x_1 and x_2 as independent variable.

Formula:

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{23}r_{13}}{1 - r_{23}^2}}$$

$$R_{2.13} = \sqrt{\frac{r_{12}^2 + r_{23}^2 - 2r_{12}r_{23}r_{13}}{1 - r_{13}^2}}$$

$$R_{3.12} = \sqrt{\frac{r_{13}^2 + r_{23}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{12}^2}}$$

Partial Correlation Coefficient

In partial correlation we study the relationship between two variables keeping the other variables as constant i.e. eliminating the effects of other variables.

Consider three variables x_1 , x_2 and x_3 then:

$r_{12.3}$ = Partial correlation coefficient between x_1 and x_2 eliminating the effect of x_3 .

$r_{23.1}$ = Partial correlation coefficient between x_2 and x_3 eliminating the effect of x_1 .

$r_{13.2}$ = Partial correlation coefficient between x_1 and x_3 eliminating the effect of x_2 .

Formula:

$$r_{12.3} = \frac{r_{12} - r_{13} \cdot r_{23}}{\sqrt{1-r_{13}^2} \sqrt{1-r_{23}^2}}$$

$$r_{23.1} = \frac{r_{23} - r_{12} \cdot r_{13}}{\sqrt{1-r_{12}^2} \sqrt{1-r_{13}^2}}$$

$$r_{13.2} = \frac{r_{13} - r_{12} \cdot r_{23}}{\sqrt{1-r_{12}^2} \sqrt{1-r_{23}^2}}$$

$$r = \frac{n \sum XY - \sum X \cdot \sum Y}{\sqrt{n \sum X^2 - (\sum X)^2} \sqrt{n \sum Y^2 - (\sum Y)^2}}$$

$$r_{12} = \frac{n \sum X_1 X_2 - \sum X_1 \cdot \sum X_2}{\sqrt{n \sum X_1^2 - (\sum X_1)^2} \sqrt{n \sum X_2^2 - (\sum X_2)^2}}$$

Q. Given:

$$r_{12} = 0.55$$

$$r_{23} = 0.61$$

$$r_{13} = 0.59$$

To find (I) $r_{12.3}$

(II) $r_{13.2}$

(III) $r_{23.1}$

$$r_{12.3} = \frac{0.55 - 0.59 \times 0.61}{\sqrt{1-0.59^2} \sqrt{1-0.61^2}}$$

=

Q. Given:

weight - X_1 , height - X_2 , Age - X_3

and $r_{12} = 0.60$, $r_{13} = 0.40$, $r_{23} = 0.50$

(I) The multiple correlation coefficient assuming weight as dependent variable.

$$\begin{aligned}
 R_{1.23} &= \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{23}r_{13}}{1 - r_{23}^2}} \\
 &= \sqrt{\frac{(0.6)^2 + (0.4)^2 - 2 \times 0.6 \times 0.4 \times 0.5}{1 - (0.5)^2}} \\
 &= 0.611 //
 \end{aligned}$$

(II) The multiple correlation coefficient assuming height as dependent variable.

Here;

$$\begin{aligned}
 R_{2.13} &= \sqrt{\frac{r_{12}^2 + r_{23}^2 - 2r_{12}r_{23}r_{13}}{1 - r_{13}^2}} \\
 &= \sqrt{\frac{0.6^2 + 0.5^2 - 2 \times 0.6 \times 0.5 \times 0.4}{1 - 0.4^2}} \\
 &= 0.6636 //
 \end{aligned}$$

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